

Data Cleaning Pipeline – StatCan Fix & Validation Report

Objective

Investigated and fixed duplicate-grain issues in the StatCan data-cleaning pipeline, specifically within `clean_statcan_long`.

Issue Identified

The original `clean_statcan_long` function was producing significant duplicate records across several StatCan output files. The most severe issue was found in `stress_coping.csv`, which contained:

- **14,388 rows**
- **11,510 duplicate grain records**

The function existed in two locations: 1. `02_data_cleaning.ipynb` — Cell 8 2. `02_data_cleaning_functions.py`

The notebook was using its own in-notebook version of the function, meaning changes made only to the `.py` file would not affect the pipeline execution.

Work Completed

A dedicated **§4·FIX** — `clean_statcan_long` section was added to `02_data_cleaning.ipynb`.

The fix includes:

- A corrected version of `clean_statcan_long`
- A self-check loop to validate duplicate grain counts
- Documentation identifying the two root causes
- Placement of the fix after the original §4 function definition so that the corrected function overrides the broken version during execution

The notebook was then restarted and the complete pipeline was run again.

Validation Results

StatCan File	Before	After	Duplicate Grain After
<code>stress_coping.csv</code>	14,388 rows	3,220 rows	0
<code>suicidal_thoughts.csv</code>	4,994 rows	1,104 rows	0

StatCan File	Before	After	Duplicate Grain After
<code>perceived_health_quarterly.csv</code>	6,318 rows	1,053 rows	0
<code>cchs_mh_disorders.csv</code>	74,036 rows	12,917 rows	0
<code>perceived_mh_annual.csv</code>	936 rows	936 rows	0

`stress_coping.csv` Validation

The corrected output was further checked and confirmed to have:

- **3,220 rows**
- **0 duplicate grain records**
- **11 clean columns**
- No null values
- Values within their corresponding confidence intervals
- Data covering **11 provinces**
- Data covering **2016 and 2019**
- **10 indicators**
- `quality_flag = E` retained for the 952 small-cell records rather than being incorrectly removed

Pipeline Result

The complete notebook was executed successfully with **0 errors**, and the cleaned outputs were regenerated in:

`data/processed/02_cleaned/`

Follow-Up

The `.py` function file was intentionally not modified as part of the final implementation. The notebook currently contains the working patch.

The recommended next step is for the team to review the fix and then synchronize the corrected function into both:

- `02_data_cleaning.ipynb`
- `02_data_cleaning_functions.py`

After synchronization, the temporary **\$4-FIX** cell can be removed to keep the notebook clean and maintain a single source of truth for the cleaning logic.